

How Do AirTags Work 🍏

#64: A Simple Introduction to AirTags Architecture (3 Minutes)



NEO KIM

JAN 08, 2025 • PAID

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This post outlines how AirTag works. You will find references at the bottom of this page if you want to go deeper.

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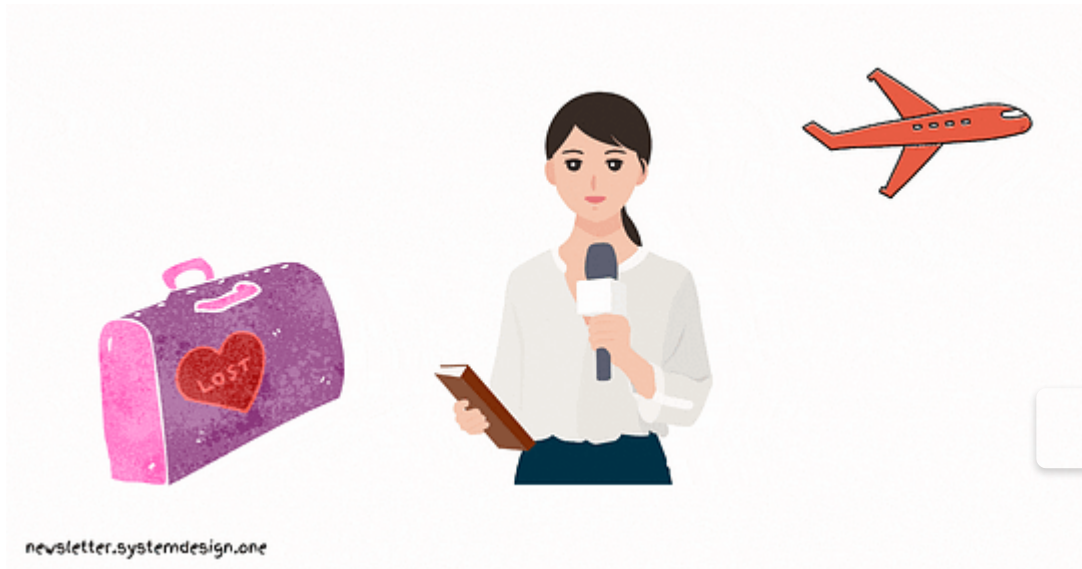
Note: This post is based on my research and may differ from real-world implementation.

Once upon a time, there was a sports journalist named Maria.

She had to travel often by flight for work.

But one day she lost her luggage during a flight, and couldn't track it.

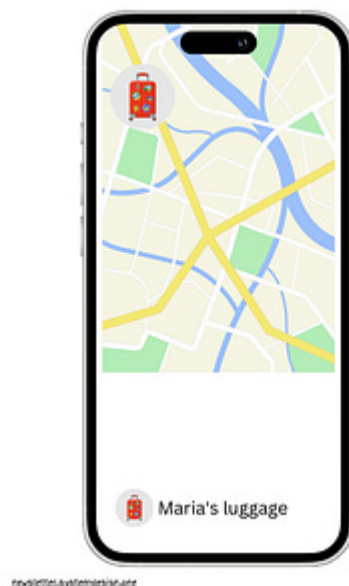
So she was sad and upset.



She hears about a tracking device, called Apple AirTag, from a coworker.

And bought it immediately, she was dazzled by its simplicity in tracking luggage.

An AirTag contains a low-power CPU and a tiny amount of memory.



A tracking device must send its accurate location data at regular intervals.

Yet finding location data using GPS, WiFi, or cellular networks consumes power. Besides it's expensive to maintain a tracking device with GPS functionality.

So smart engineers at Apple used simple ideas to solve this hard problem.

Onward.



How Do AirTags Work

An AirTag doesn't use GPS, WiFi, or a cellular network for communication; instead, it uses Bluetooth Low Energy (**BLE**).

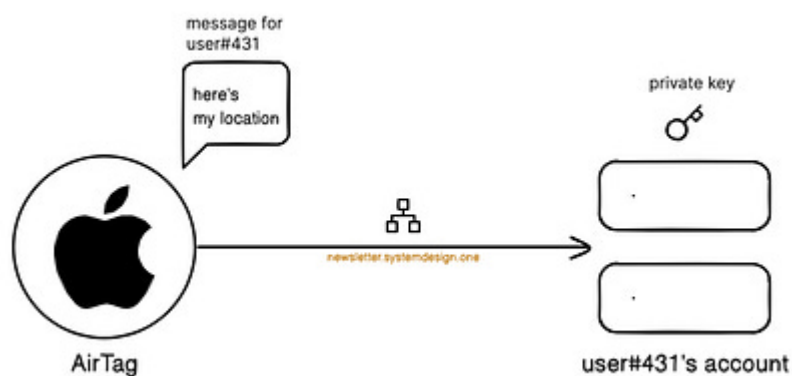
Imagine BLE as a low-power wireless technology for communication.

Here's how AirTag works:

1. Setting up AirTag

They generate a public-private key pair, using elliptic curve cryptography, when a user adds an AirTag.

The key pair is shared between AirTag and the user account. The AirTag sends location data after encrypting it using the public key. While the user account decrypts the received location data using the private key.



Sending Location Data With End-To-End Encryption

Think of the **public key** as an email address; anyone can send messages to it. While the **private key** is similar to the password of an inbox; only the user with the password can read emails.

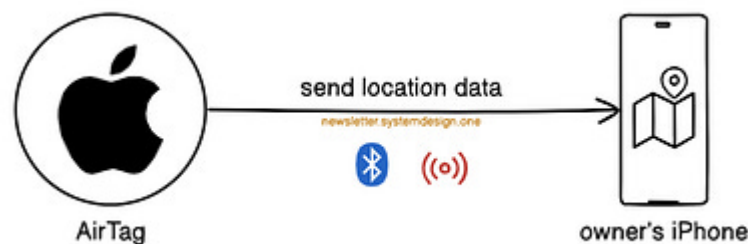
And authenticity of an email can be verified by checking the sender's email address: **digital signature**.

Ready for the best part?

2. Broadcasting AirTag's Location

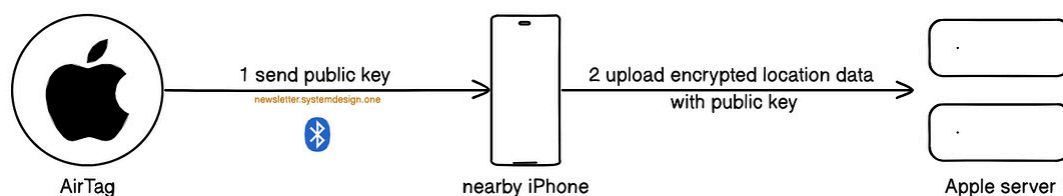
They send the AirTag's location, which is near its owner's iPhone, using Bluetooth, or Ultra Wideband for precision and efficiency.

Think of **Ultra Wideband** as a wireless technology for high-speed, and short-range communication.



Sending Location Data of an AirTag Using Bluetooth or Ultra Wideband When Owner's iPhone Is Nearby

Yet Bluetooth and Ultra Wideband communication won't work if the owner's iPhone is far from the AirTag. So they rely on someone else's iPhone which is nearby.



AirTag Sending Location Data When Owner's iPhone Is Far

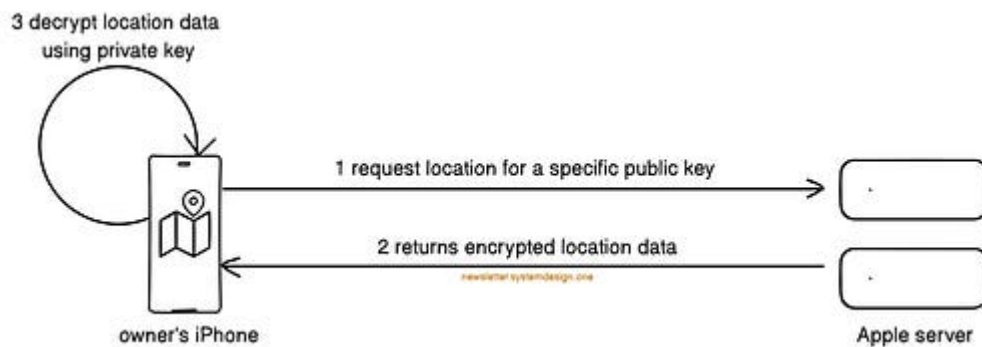
Here's how it works:

- The AirTag *broadcasts its public key every 2 seconds over BLE*.
- Someone else's iPhone, which is nearby, receives the broadcast signal.
- The iPhone encrypts its location data and timestamp using the received public key.
- The iPhone uploads the encrypted data to the Apple server over HTTP.

Put simply, AirTag doesn't send location data; instead, only the public key. The iPhone then includes its location data and encrypts it.

3. Finding AirTag's Location

The owner can find AirTag's location using their iPhone app.



The Owner Decrypting Received Location Data Using Private Key

Here's how it works:

- The owner authenticates on the Apple server.
- The owner requests location data for a specific public key.
- The owner *decrypts the received location data using their private key*.

The iPhone then displays AirTag's recent location on a map.

An iPhone is necessary to relay location data, so an AirTag won't work if there aren't any iPhones nearby.

Put simply, the precision and effectiveness of an AirTag depends on the number of iPhones in its proximity.

While there are around 1.46 billion iPhone users in the world.



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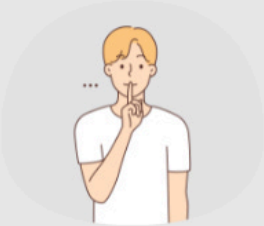
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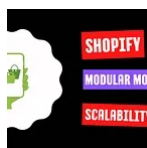
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References

- [Using Find My to locate missing Apple devices](#)
- [Track You: A Deep Dive into Safety Alerts for Apple AirTags](#)
- [Apple AirTag Reverse Engineering](#)
- [Dissecting the Encryption Protocols Inside Apple AirTags](#)
- [How does the AirTag work?](#)
- [What is a Beacon? A Guide to Bluetooth Beacon Technology](#)
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